MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

GATE, NAND, DIGITAL, KEY CODE LDQ

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic NAND gate module.

1.2 Classification. This specification covers two classes of modules.

Class 1 and class 11 as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

*2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

*2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:28Z
Check the source to verify that this is the current version before use.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.18 failures per million operating hours for class I or 0.45 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.4).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.4).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be LDQ.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/18-1 for class I and M28787/18-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:28Z
Check the source to verify that this is the current version before use.

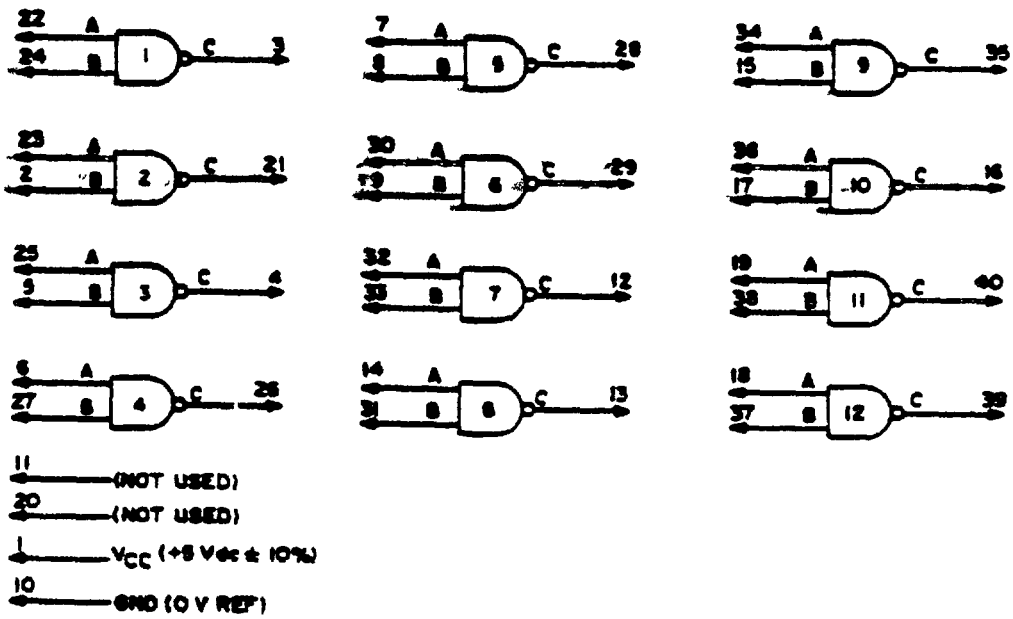


FIGURE 1. LDQ module function diagram, NAND gate.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:28Z
Check the source to verify that this is the current version before use.

*TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5°C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1-A12	V_{OUT} (low loaded)	III	S	0.01	0.80	0.01	0.80	0.01	0.80	0.01	0.80	0.01	0.80	V
B1-B24	Input breakdown current	IV	S	0	1	0	1	0	1	0	1	0	1	mA
C1-C24	Input clamp diode voltage	V	S	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
D1-D12	Output voltage, low level	VI	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
E1-E24	Output voltage, high level	VII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
F1-F24	Input current, low level	VIII	X	0.40	1.73	0.40	1.73	0.40	1.73	0.40	1.73	0.40	1.73	mA
G1-G24	Input current, high level	IX	X	0	30	0	40	0	40	0	40	0	40	µA
H1-H12	Output short circuit current	X	X	20	55	20	55	20	55	20	55	20	55	mA
J1	Power supply current, low level	XI	S	1	66	1	66	1	66	1	66	1	66	mA
J2	Power supply current, high level	XI	S	1	24	1	24	1	24	1	24	1	24	mA
K1-K12	Transition time, (t_{TLH})	XII	S	4	45	4	50	4	50	4	50	4	55	ns
L1-L12	Transition time, (t_{THL})	XII	S	4	20	4	25	4	25	4	25	4	30	ns
M1-M12	Propagation delay time (t_{PHL})	XIII	S	4	19	4	22	4	22	4	22	4	25	ns
N1-N12	Propagation delay time (t_{PLH})	XIII	S	4	28	4	30	4	30	4	30	4	32	ns
P1-P4, P9-P12	Output crosstalk, low level	XIV	Q	0	610	0	610	0	610	0	610	0	610	mV
P5-P8	Output crosstalk, low level	XIV	Q	0	550	0	550	0	550	0	550	0	550	mV
P13-P24	Output crosstalk, low level	XIV	Q	0	280	0	280	0	280	0	280	0	280	mV
R1-R12	Output crosstalk, high level	XV	Q	0	400	0	400	0	400	0	400	0	400	mV
S	Power supply transient amplitude	XVI	Q	0	400	0	400	0	400	0	400	0	400	mV
T	Power supply transient rise time	XVII	S	1	2	1	2	1	2	1	2	1	2	ms

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:28Z
Check the source to verify that this is the current version before use.

*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5°C		-65°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
U1	Isolation													
	Initial	XVIII	S	10	-	10	-	10	-	10	-	10	-	MΩ
	End-of-life	XVIII	S	1	-	1	-	1	-	1	-	1	-	MΩ
U2,U3	Isolation, unused pins													
	Initial	XVIII	Q	10	-	10	-	10	-	10	-	10	-	MΩ
	End-of-life	XVIII	Q	1	-	1	-	1	-	1	-	1	-	MΩ
V1-V24	Input capacitance	XIX	Q	2	15	2	15	2	15	2	15	2	15	pF
W1-W24	Output impedance (power off) check	XX	Q	3.50	5.55	3.50	5.55	3.50	5.55	3.50	5.55	3.50	5.55	V

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

Source: https://assist.dla.mil -- Downloaded: 2026/0-20T15:28Z
Check the source to verify that this is the current version before use.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through W shall be performed in any order with the following exceptions. If test conditions A, B or U are part of a sequence of tests, the order of testing shall be as follows: A, B, U where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through W in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C through W and T shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition T, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests C through W and T shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests C through W shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Military Specification MIL-M-28787/18 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No. 10001-2833408. Part number M28787/18-1 supersedes part numbers 2833408 and M28787/18.

6.4 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:28Z
Check the source to verify that this is the current version before use.

*6.5 Module description. The following electrical parameters reflect the 0°C to 60°C and -55°C to 85°C end-of-life limits from table I.

Maximum power dissipation	0.363 W (all outputs in low level)
Input voltage	
Low level	0.8 V maximum
High level	2.0 V minimum
Input current	
Low level (V_{IL} = 0 V)	1.73 mA maximum
High level (V_{IH} = 2.4 V)	40 µA maximum
Output voltage	
Low level	0.4 V maximum
High level	2.4 V minimum, V_{CC} maximum
Output drive	
Low level (V_{OL} = 0.4 V)	16 mA maximum
High level (V_{OH} = 2.4 V)	400 µA maximum
Propagation delay time	
t_{PLH}	28 ns maximum
t_{PHL}	22 ns maximum

6.6 Module Function. This module contains twelve independent 2-input NAND gates as shown on figure 1.

6.6.1 Input to output relationship. Each NAND gate has the input to output relationship of table II for positive logic.

Table II. Input to output relationship.

Input		Output
A	B	C
L	L	H
L	H	H
H	L	H
H	H	L



$C = \overline{AB}$

6.6.2 Parallel outputs. Paralleling of outputs is not permissible except when both inputs and outputs of circuits are paralleled to obtain extended drive. Each circuit whose output is paralleled with an output of another circuit must have each of its inputs paralleled with a corresponding input of the other circuit. Circuits may be paralleled only with other circuits within the same group. These circuit groups are:

a. Circuits 1, 2, 3, and 4

b. Circuits 5, 6, 7, and 8

c. Circuits 9, 10, 11, and 12.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:28Z
Check the source to verify that this is the current version before use.

The output drive capability of this combination is equal to the sum of the output drive capabilities of the individual circuits. The input loading of this combination is equal to the sum of the input loading of the individual circuits.

6.6.3 Power supply voltage. The module shall conform to the requirements of table I for a power supply voltage of 5.0 V dc ±10 percent when driving the specified loads in either logic state except that the limits for output short circuit current and output impedance apply only for a power supply voltage of 5.5 V dc ± 1 percent.

6.6.4 Crosstalk relationships. The module has the following crosstalk relationships.

6.6.4.1 Module crosstalk. When connected in the worst case mode of operation (all but one gate switching simultaneously from a high level to a low level), the LDQ module could exhibit substantial crosstalk. When the quiet line is a low level, there may be as much as a 610 mV positive going spike coincident with the negative edge of the switching output.

6.6.4.2 Circuit crosstalk. The crosstalk on any output is nearly directly proportional to the number of switching outputs. For every switching output:

a. In group A (circuits 1 thru 4) or group C (circuits 9 thru 12) the noise induced on a low level quiet output in that group is as much as 110 mV.

b. In group B (circuits 5 thru 8) the noise induced on a low level quiet output in that group is as much as 90 mV.

c. Outside a particular group the noise induced on a low level quiet output in that particular group is as much as 35 mV.

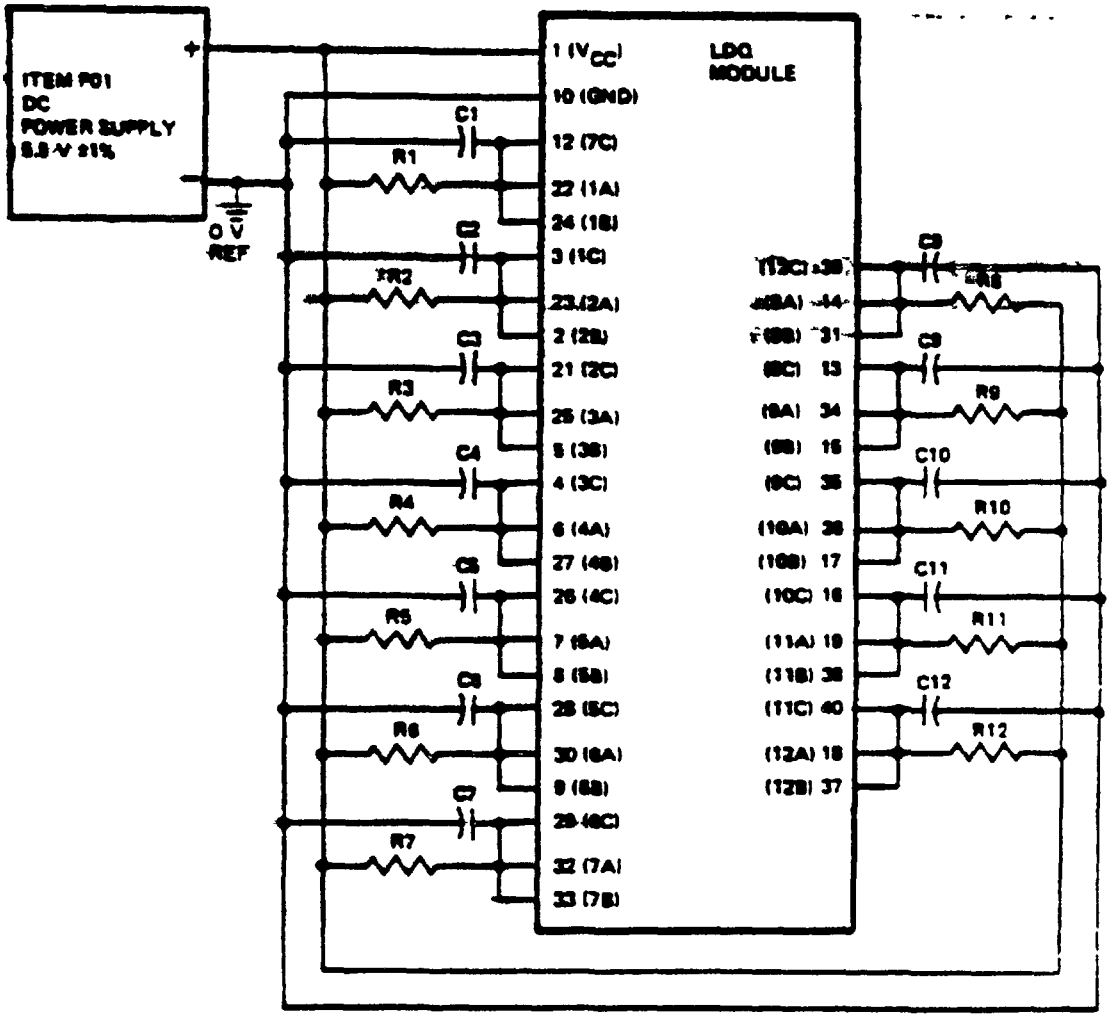
6.7 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5963-N457)

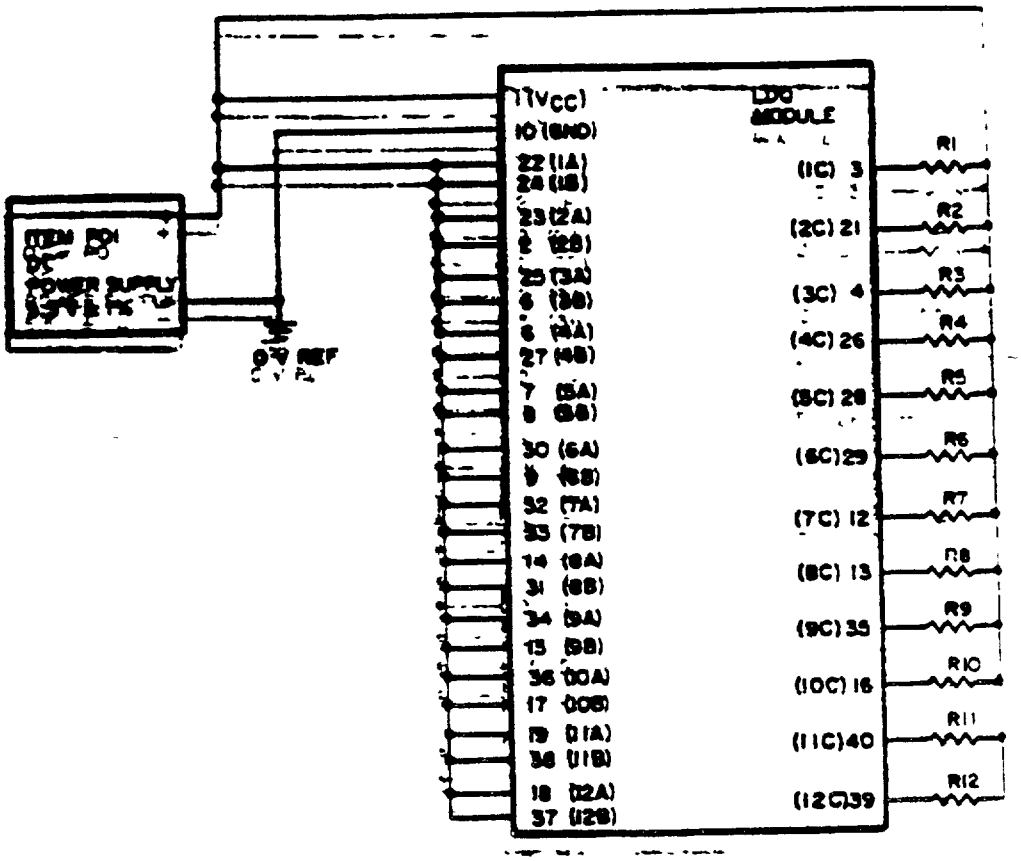
Source: https://assist.dla.mil -- Downloaded: 202610-20T15:28Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1-R12 = 316Ω ±1%, 1/4 W.
2. C1-C12 = 100 pF ±20%.
3. When using figure 2 for module burn-in, the RC loads are not required.

*FIGURE 2. Operating temperature, life, and burn-in test circuit.

Source: https://assist.dla.mil -- Downloaded: 202640-20T15:28Z
Check the source to verify that this is the current version before use.



NOTE R1-R12 = 316Ω ±1%, 1/4 W

FIGURE 3. Component temperature test circuit

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:28Z
Check the source to verify that this is the current version before use.

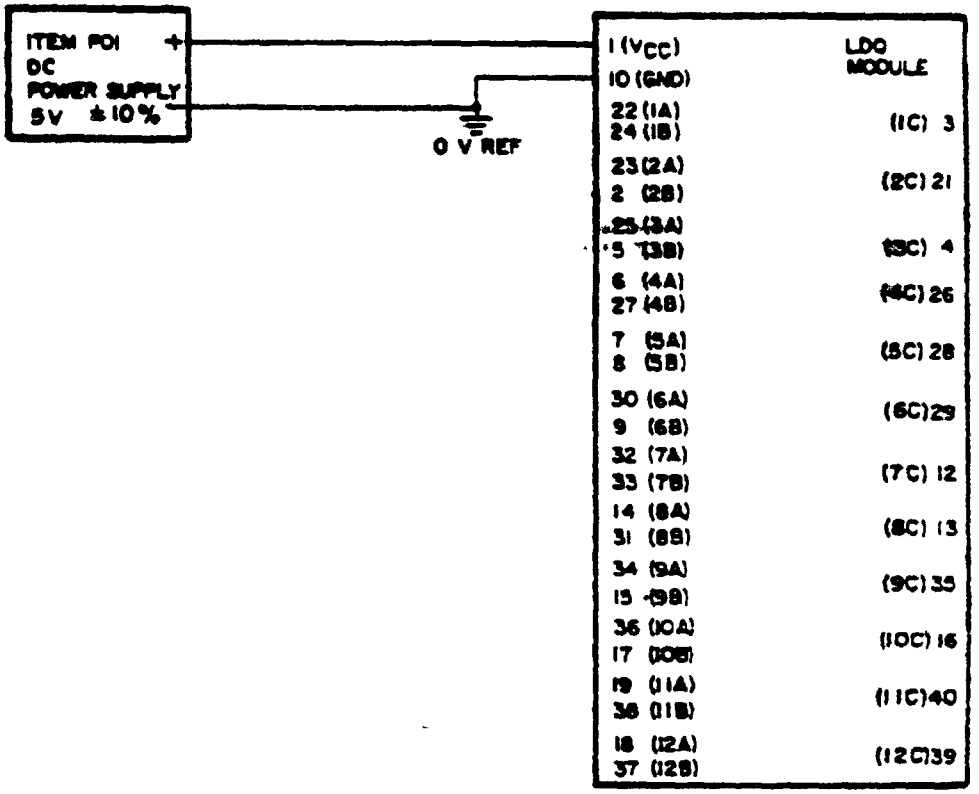


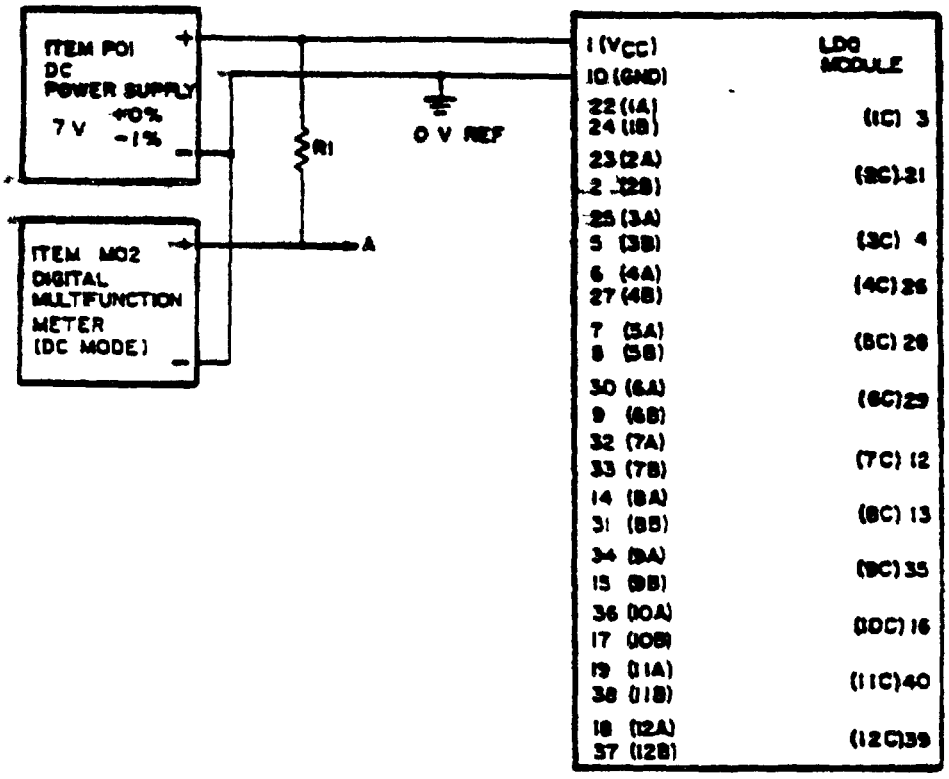
FIGURE 4. Vibration test circuit

Source: https://assist.dla.mil -- Downloaded: 202640-20T15:28Z
Check the source to verify that this is the current version before use.

TABLE III. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect: A to pin	Initial limits 25°C ambient: Min	Initial limits 25°C ambient: Max	Unit	Test figure
		3	0.01	0.08	V	5
A1	1C V_{OUT} (low loaded)	21	•	•	•	•
A2	2C	4	•	•	•	•
A3	3C	20	•	•	•	•
A4	4C	19	•	•	•	•
A5	5C	22	•	•	•	•
A6	6C	12	•	•	•	•
A7	7C	13	•	•	•	•
A8	8C	16	•	•	•	•
A9	9C	15	•	•	•	•
A10	10C	18	•	•	•	•
A11	11C	17	•	•	•	•
A12	12C	14	•	•	•	•

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:28Z
Check the source to verify that this is the current version before use.



R1 = 316Ω ±1%, 1/4 W.

FIGURE 5. V_{OUT} (low-loaded) test circuit for table III.

Source: https://assist.dla.mil -- Downloaded: 202640-20T15:28Z
Check the source to verify that this is the current version before use.

TABLE IV. Input breakdown current test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	1A Input breakdown current	22	24	0	1	mA	6
B2	1B "	24	22	"	"	"	"
B3	2B "	2	23	"	"	"	"
B4	2A "	23	2	"	"	"	"
B5	3B "	5	25	"	"	"	"
B6	3A "	25	5	"	"	"	"
B7	4A "	6	27	"	"	"	"
B8	4B "	27	6	"	"	"	"
B9	5A "	7	8	"	"	"	"
B10	5B "	8	7	"	"	"	"
B11	6B "	9	30	"	"	"	"
B12	6A "	30	9	"	"	"	"
B13	7A "	32	33	"	"	"	"
B14	7B "	33	32	"	"	"	"
B15	8A "	14	31	"	"	"	"
B16	8B "	31	14	"	"	"	"
B17	9B "	15	34	"	"	"	"
B18	9A "	34	15	"	"	"	"
B19	10B "	17	36	"	"	"	"
B20	10A "	36	17	"	"	"	"
B21	11A "	19	38	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:28Z
Check the source to verify that this is the current version before use.

TABLE IV. Input breakdown current test - Continued.

Test condition	Parameter		Connect		Internal limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
022	11B	Input breakdown current	36	19	0	1	mA	6
023	12A	"	18	37	"	"	"	"
024	12B	"	37	18	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:28Z
Check the source to verify that this is the current version before use.

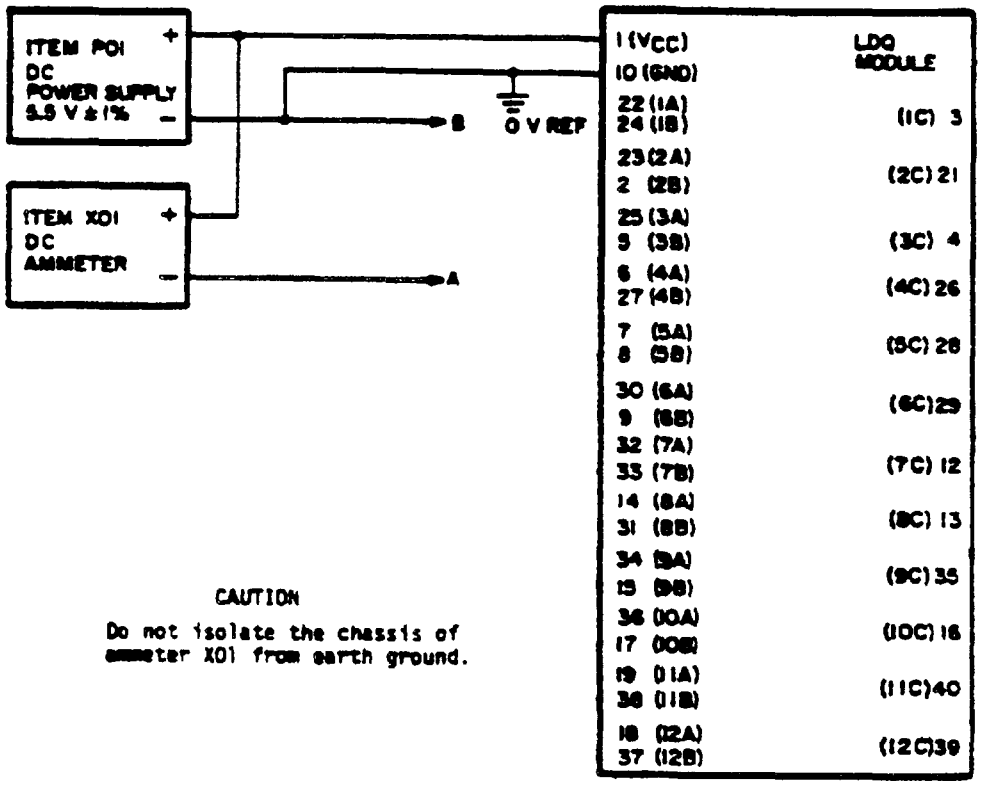


FIGURE 6. Input breakdown current test circuit for table IV

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:28Z
Check the source to verify that this is the current version before use.

TABLE V. Input clamp diode voltage test.

Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
C1	1A Input clamp diode voltage	22	-0.4	-1.5	V	7
C2	1B "	24	"	"	"	"
C3	2B "	2	"	"	"	"
C4	2A "	23	"	"	"	"
C5	3B "	5	"	"	"	"
C6	3A "	25	"	"	"	"
C7	4A "	6	"	"	"	"
C8	4B "	27	"	"	"	"
C9	5A "	7	"	"	"	"
C10	5B "	8	"	"	"	"
C11	6B "	9	"	"	"	"
C12	6A "	30	"	"	"	"
C13	7A "	32	"	"	"	"
C14	7B "	33	"	"	"	"
C15	8A "	14	"	"	"	"
C16	8B "	31	"	"	"	"
C17	9B "	15	"	"	"	"
C18	9A "	34	"	"	"	"
C19	10B "	17	"	"	"	"
C20	10A "	36	"	"	"	"
C21	11A "	19	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 20260-20T15:28Z
Check the source to verify that this is the current version before use.

TABLE V. Input clamp diode voltage test - Continued.

Test condition	Parameter		Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
C22	11B	Input clamp diode voltage	38	-0.4	-1.5	V	7
C23	12A	"	18	"	"	"	"
C24	12B	"	37	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 202640-20T15:28Z
Check the source to verify that this is the current version before use.

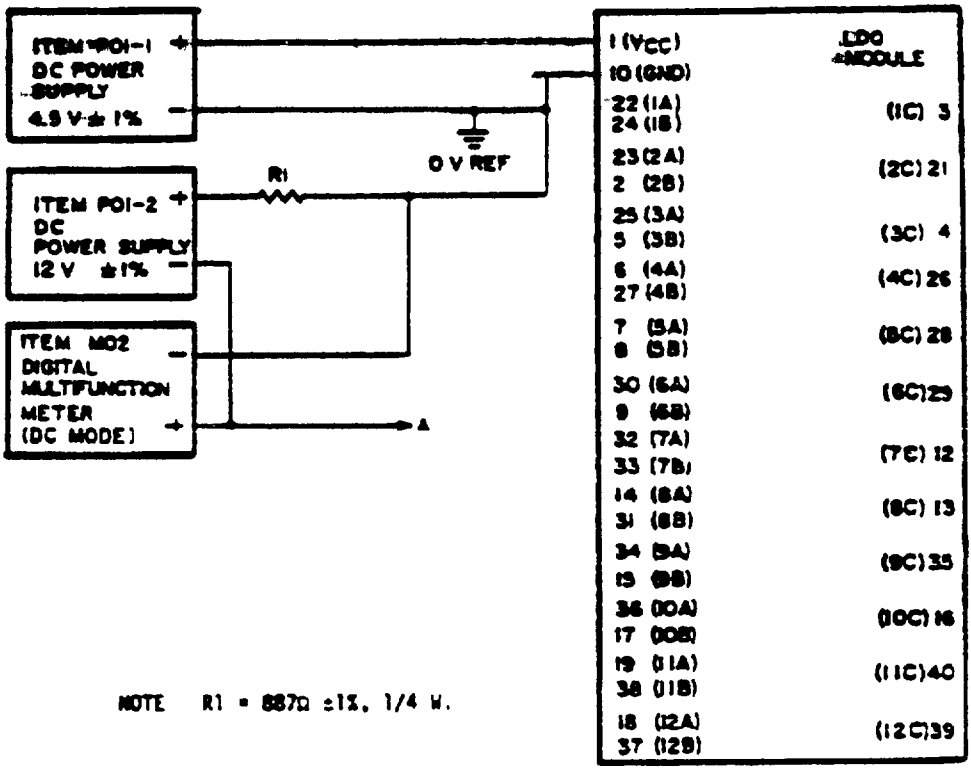


FIGURE 7. Input clamp diode voltage test circuit for table V.

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:28Z
Check the source to verify that this is the current version before use.

TABLE VI. Output voltage (low-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
D1	1C Output voltage, low level	22,24	Open	3	0.0	0.4	V	8
D2	2C "	2,23	"	21	"	"	"	"
D3	3C "	5,25	"	4	"	"	"	"
D4	4C "	6,27	"	26	"	"	"	"
D5	5C "	7,8	"	28	"	"	"	"
D6	6C "	9,30	"	29	"	"	"	"
D7	7C "	32,33	"	12	"	"	"	"
D8	8C "	14,31	"	13	"	"	"	"
D9	9C "	15,34	"	35	"	"	"	"
D10	10C "	17,36	"	16	"	"	"	"
D11	11C "	19,38	"	40	"	"	"	"
D12	12C "	18,37	"	39	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T15:28Z
Check the source to verify that this is the current version before use.

TABLE VII. Output voltage (high-level) test.

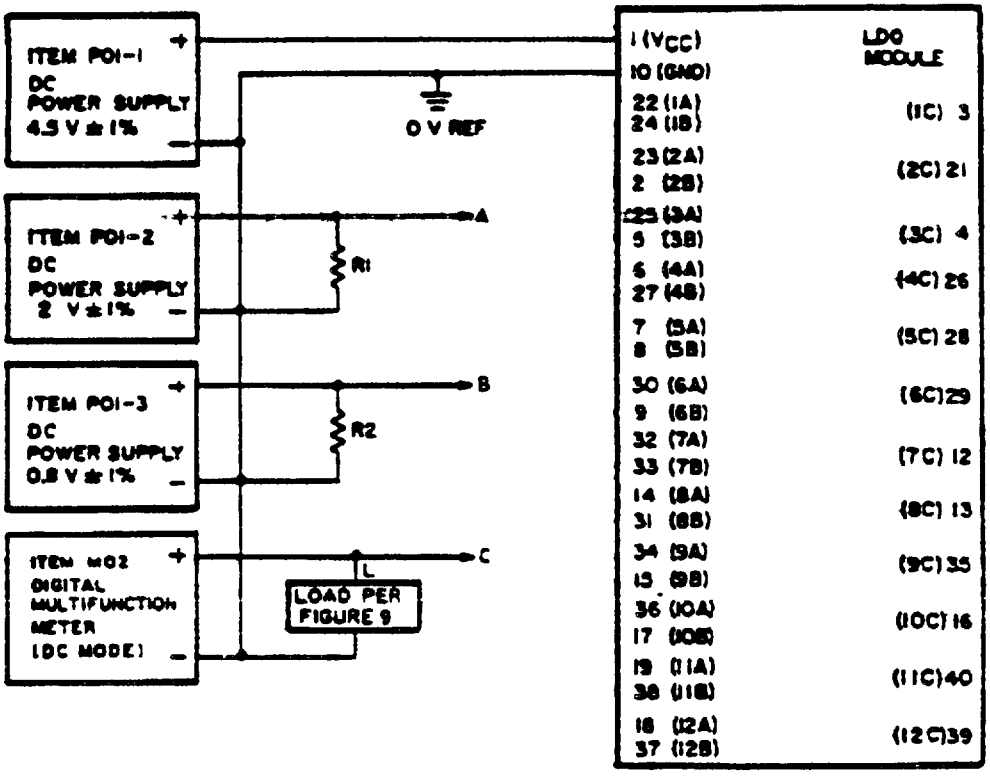
Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
E1	1C Output voltage, high level	Open	22	3	2.4	4.5	V	8
E2	1C "	"	24	3	"	"	"	"
E3	2C "	"	2	21	"	"	"	"
E4	2C "	"	23	21	"	"	"	"
E5	3C "	"	5	4	"	"	"	"
E6	3C "	"	25	4	"	"	"	"
E7	4C "	"	6	26	"	"	"	"
E8	4C "	"	27	26	"	"	"	"
E9	5C "	"	7	28	"	"	"	"
E10	5C "	"	8	28	"	"	"	"
E11	6C "	"	9	29	"	"	"	"
E12	6C "	"	30	29	"	"	"	"
E13	7C "	"	32	12	"	"	"	"
E14	7C "	"	33	12	"	"	"	"
E15	8C "	"	14	13	"	"	"	"
E16	8C "	"	31	13	"	"	"	"
E17	9C "	"	15	35	"	"	"	"
E18	9C "	"	34	35	"	"	"	"
E19	10C "	"	17	16	"	"	"	"
E20	10C "	"	36	16	"	"	"	"
E21	11C "	"	19	40	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:28Z
Check the source to verify that this is the current version before use.

TABLE VII. Output voltage (high-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
E22	11C Output voltage, high level	Open	38	40	2.4	4.5	V	8
E23	12C "	"	18	39	"	"	"	"
E24	12C "	"	37	39	"	"	"	"

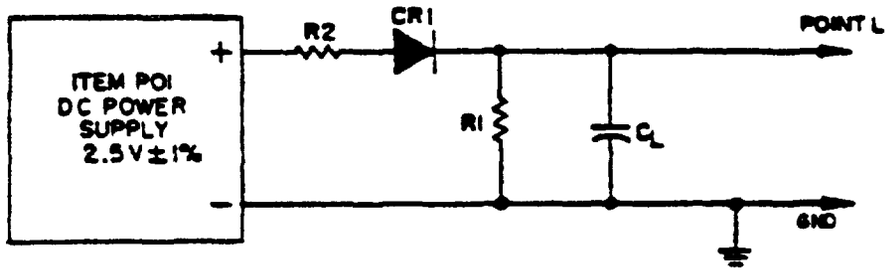
Source: https://assist.dla.mil -- Downloaded: 2026-02-20T15:28Z
Check the source to verify that this is the current version before use.



NOTE. R1 = R2 = 51Ω ±5%, 1 W.

FIGURE 8. Output voltage (low-level and high-level) test circuit for tables VI and VII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:28Z
Check the source to verify that this is the current version before use.



NOTES
1. C_L = 100 ±5 pF (including probe and parasitic capacitance) for test conditions K, L, M and N.
2. CR1 = 1N4150 or equivalent.
3. R1 = 6.04 kΩ ±1%, 1/8 W.
4. R2 = select value for a current flow of 16 mA ±1% with point "L" held at 0.4 V.
5. C_L is not required for dc electrical tests D1-D12 or E1-E24.
6. The maximum lead length from point "L" to the appropriate module pin shall be 3 inches (76.2 mm).
7. During output switching, the 2.5 V power supply shall have a maximum overshoot of 500 mV
8. C_L = 50 ±5 pF (including probe and parasitic capacitance) for test conditions P and R.

* FIGURE 9. LDQ logic load circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:28Z
Check the source to verify that this is the current version before use.

TABLE VIII. Input current (low-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
F1	1B Input current, low level	Open	24	Open	0.40	1.73	mA	10
F2	1A "	"	22	"	"	"	"	"
F3	2A "	"	23	"	"	"	"	"
F4	2B "	"	2	"	"	"	"	"
F5	3A "	"	25	"	"	"	"	"
F6	3B "	"	5	"	"	"	"	"
F7	4B "	"	27	"	"	"	"	"
F8	4A "	"	6	"	"	"	"	"
F9	5B "	"	8	"	"	"	"	"
F10	5A "	"	7	"	"	"	"	"
F11	6A "	"	30	"	"	"	"	"
F12	6B "	"	9	"	"	"	"	"
F13	7A "	"	32	"	"	"	"	"
F14	7B "	"	33	"	"	"	"	"
F15	8B "	"	31	"	"	"	"	"
F16	8A "	"	14	"	"	"	"	"
F17	9A "	"	34	"	"	"	"	"
F18	9B "	"	15	"	"	"	"	"
F19	10A "	"	36	"	"	"	"	"
F20	10B "	"	17	"	"	"	"	"
F21	11B "	"	38	"	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026/0-20T15:28Z
Check the source to verify that this is the current version before use.

TABLE VIII. Input current (low-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
F22	11A Input current, low level	Open	19	Open	0.40	1.73	mA	10
F23	12B "	"	37	"	"	"	"	"
F24	12A "	"	18	"	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T15:28Z
Check the source to verify that this is the current version before use.

TABLE IX. Input current (high-level) test.

Test condition	Parameter		Connect			Limits 25°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
	1B	Input current, high level	22	Open	24	0	30	μA	10
G1	1A		24		22				
G2	2A		2		23				
G3	2B		23		2				
G4	3A		5		25				
G5	3B		25		5				
G6	4B		6		27				
G7	4A		27		6				
G8	5B		7		8				
G9	5A		8		7				
G10	6A		9		30				
G11	6B		30		9				
G12	7B		32		33				
G13	7A		33		32				
G14	8B		14		31				
G15	8A		31		14				
G16	9A		15		34				
G17	9B		34		15				
G18	10B		17		36				
G19	10A		36		17				
G20	11B		19		38				
G21	11A								

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:28Z
Check the source to verify that this is the current version before use.

TABLE IX. Input current (high-level) test - Continued.

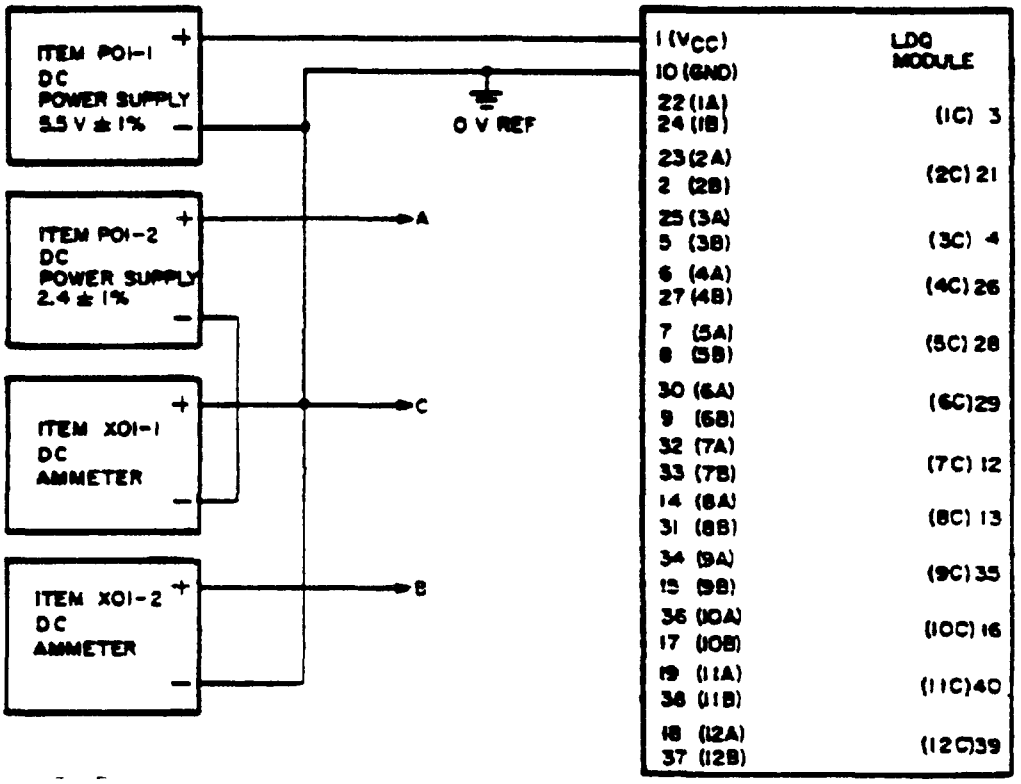
Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G22	11A Input current, high level	38	Open	19	0	30	µA	10
G23	12B "	18	"	37	"	"	"	"
G24	12A "	37	"	18	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 202640-20T15:28Z
Check the source to verify that this is the current version before use.

TABLE X. Output short circuit current test.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25°C ambient Min	Initial limits 25°C ambient Max	Unit	Test figure
		Open						10
H1	1C Output short circuit current	•	3	22	20	55	mA	•
H2	2C	•	21	2	•	•	•	•
H3	3C	•	4	5	•	•	•	•
H4	4C	•	26	6	•	•	•	•
H5	5C	•	28	7	•	•	•	•
H6	6C	•	29	9	•	•	•	•
H7	7C	•	12	32	•	•	•	•
H8	8C	•	13	14	•	•	•	•
H9	9C	•	35	15	•	•	•	•
H10	10C	•	16	17	•	•	•	•
H11	11C	•	40	19	•	•	•	•
H12	12C	•	39	18	•	•	•	•

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:28Z
Check the source to verify that this is the current version before use.



CAUTION

Do not isolate the chassis of ammeter XO1 from earth ground

NOTES.
1. Since the items XO1-1 and XO1-2 are not used simultaneously, only one item XO1 is required. Use item XO1-2 for tables VIII and X, use item XO1-1 for table IX.
2. With point "A" disconnected, item XO1-1 should indicate zero current on the 1 µA range
3. When performing output short circuit tests, only one output shall be shorted at a time and the measurement taken within 5 seconds.

FIGURE 10 Input current (low-level and high-level) and output short circuit current test circuit for tables VIII, IX and X

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:28Z
Check the source to verify that this is the current version before use.

TABLE II. Power supply current (low-level and high-level) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J1	Power supply current, low level	Open	Open	1	66	mA	11
J2	Power supply current, high level	B	A	1	24	mA	11

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:28Z
Check the source to verify that this is the current version before use.

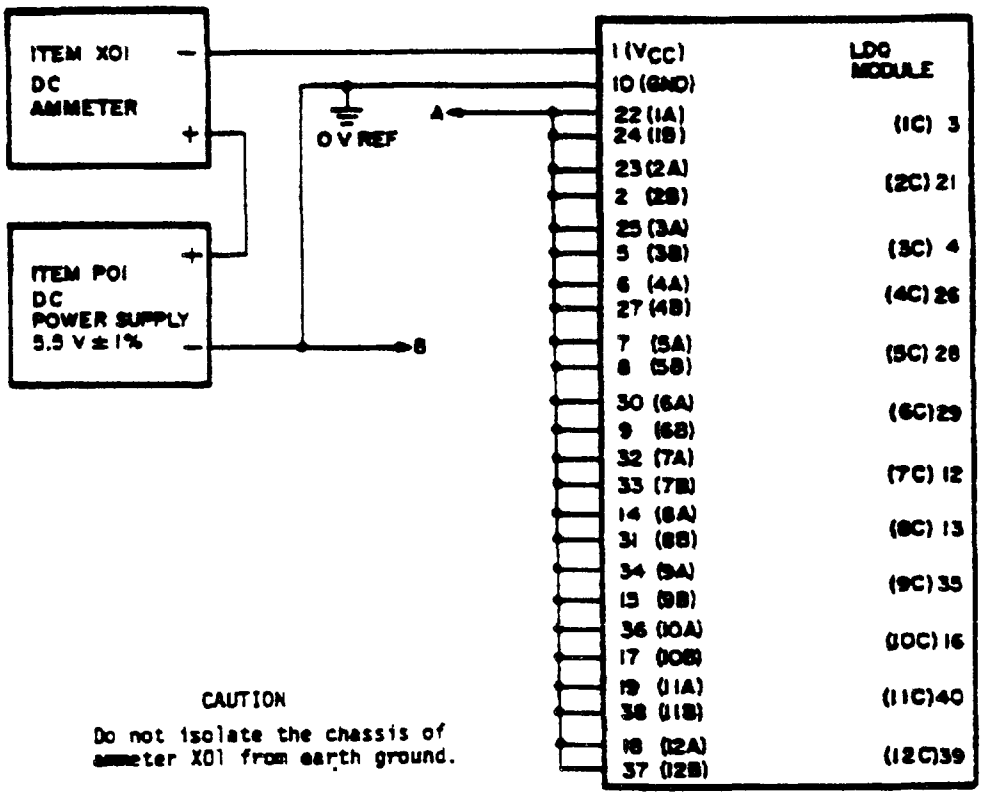


FIGURE 11 Power supply current (low-level and high-level) test circuit for table XI

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:28Z
Check the source to verify that this is the current version before use.

TABLE XII. Transition time (t_{TLH} and t_{THL}) test.

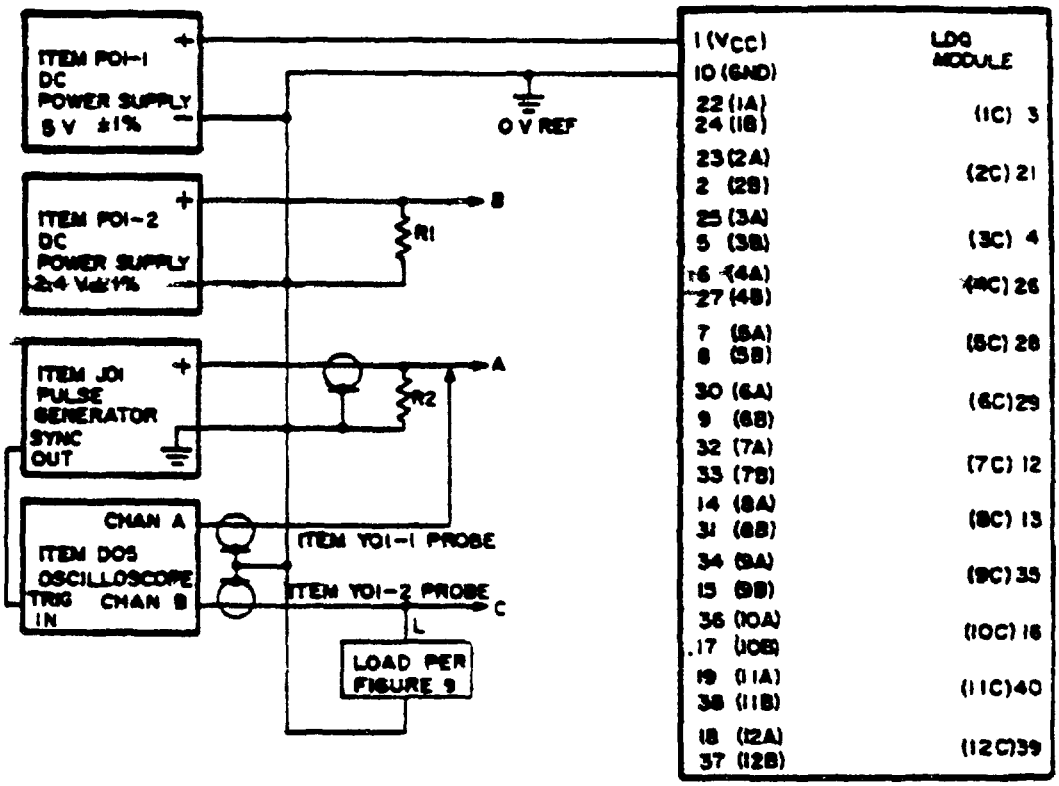
Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient				Unit	Test figure
		A to pin	B to pin	C to pin	t_{TLH}		t_{THL}			
					Min	Max	Min	Max		
K1,L1	1C Transition time	22	24	3	4	45	4	20	ns	12
K2,L2	2C "	2	23	21	"	"	"	"	"	"
K3,L3	3C "	5	25	4	"	"	"	"	"	"
K4,L4	4C "	6	27	26	"	"	"	"	"	"
K5,L5	5C "	7	8	28	"	"	"	"	"	"
K6,L6	6C "	9	30	29	"	"	"	"	"	"
K7,L7	7C "	32	33	12	"	"	"	"	"	"
K8,L8	8C "	14	31	13	"	"	"	"	"	"
K9,L9	9C "	15	34	35	"	"	"	"	"	"
K10,L10	10C "	17	36	16	"	"	"	"	"	"
K11,L11	11C "	19	38	40	"	"	"	"	"	"
K12,L12	12C "	18	37	39	"	"	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:28Z
Check the source to verify that this is the current version before use.

TABLE XIII. Propagation delay time (t_{PHL} and t_{PLH}) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient				Unit	Test figure
		A to pin	B to pin	C to pin	t_{PHL}		t_{PLH}			
					Min	Max	Min	Max		
M1,N1	1C Propagation delay time	22	24	3	4	19	4	28	ns	12
M2,N2	2C "	2	23	21	"	"	"	"	"	"
M3,N3	3C "	5	25	4	"	"	"	"	"	"
M4,N4	4C "	6	27	26	"	"	"	"	"	"
M5,N5	5C "	7	8	28	"	"	"	"	"	"
M6,N6	6C "	9	30	29	"	"	"	"	"	"
M7,N7	7C "	32	33	12	"	"	"	"	"	"
M8,N8	8C "	14	31	13	"	"	"	"	"	"
M9,N9	9C "	15	34	35	"	"	"	"	"	"
M10,N10	10C "	17	36	16	"	"	"	"	"	"
M11,N11	11C "	19	38	40	"	"	"	"	"	"
M12,N12	12C "	18	37	39	"	"	"	"	"	"

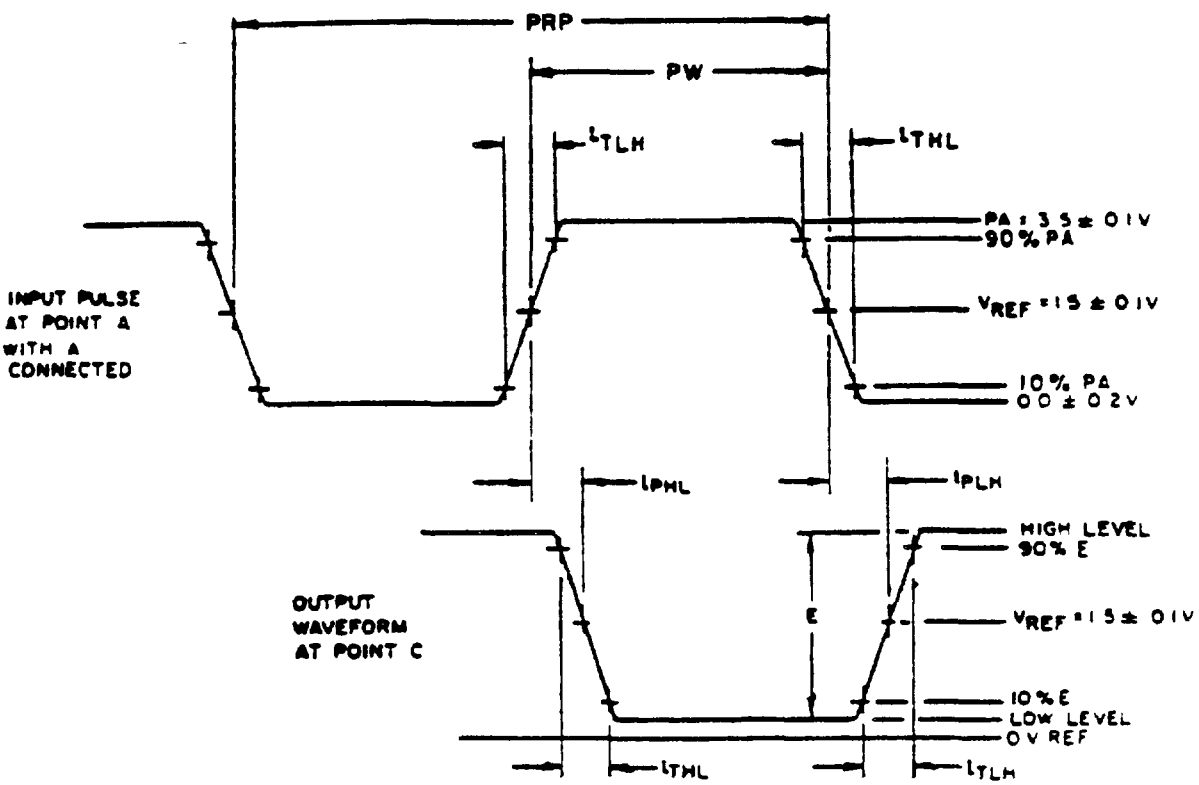
Source: https://assist.dla.mil -- Downloaded: 2026-02-20T15:28Z
Check the source to verify that this is the current version before use.



NOTE: R1 = R2 = 51Ω ±5%, 1 W.

FIGURE 12. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PHL} and t_{PLH}) test circuit and waveforms for tables XII and XIII.

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:28Z
Check the source to verify that this is the current version before use.



NOTES:
1. Input pulse parameters:
   PWA = 100 ns ±10%.
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
   PRP = 200 ns ±10%.
2. All voltage levels are referenced to ground.
3. Before making the measurements specified in tables XII and XIII, verify that the output pulse high level $\geq$ 2.4 V and the low level is $\leq$ 0.4 V.

FIGURE 12. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PHL} and t_{PLH}) test circuit and waveforms for tables XII and XIII - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:28Z
Check the source to verify that this is the current version before use.

TABLE XIV. Output crosstalk (low-level) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P1	1C Output crosstalk, low level	Open	22	6,7,14,18,19,23, 25,30,32,34,36	3	0	610	mV	13
P2	2C "	"	23	6,7,14,18,19,22, 25,30,32,34,36	21	"	"	"	"
P3	3C "	"	25	6,7,14,18,19,22, 23,30,32,34,36	4	"	"	"	"
P4	4C "	"	6	7,14,18,19,22,23, 25,30,32,34,36	26	"	"	"	"
P5	5C "	"	7	6,14,18,19,22,23, 25,30,32,34,36	28	"	590	"	"
P6	6C "	"	30	6,7,14,18,19,22, 23,25,32,34,36	29	"	"	"	"
P7	7C "	"	32	6,7,14,18,19,22, 23,25,30,34,36	12	"	"	"	"
P8	8C "	"	14	6,7,18,19,22,23, 25,30,32,34,36	13	"	"	"	"
P9	9C "	"	34	6,7,14,18,19,22, 23,25,30,32,36	35	"	610	"	"
P10	10C "	"	36	6,7,14,18,19,22, 23,25,30,32,34	16	"	"	"	"
P11	11C "	"	19	6,7,14,18,22,23, 25,30,32,34,36	40	"	"	"	"
P12	12C "	"	18	6,7,14,19,22,23, 25,30,32,34,36	39	"	"	"	"
P13	1C "	"	6,22,23,25	7,14,18,19,30, 32,34,36	3	"	280	"	"
P14	2C "	"	6,22,23,25	7,14,18,19,30, 32,34,36	21	"	280	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T15:28Z
Check the source to verify that this is the current version before use.

TABLE XIV. Output crosstalk (low-level) test - Continued.

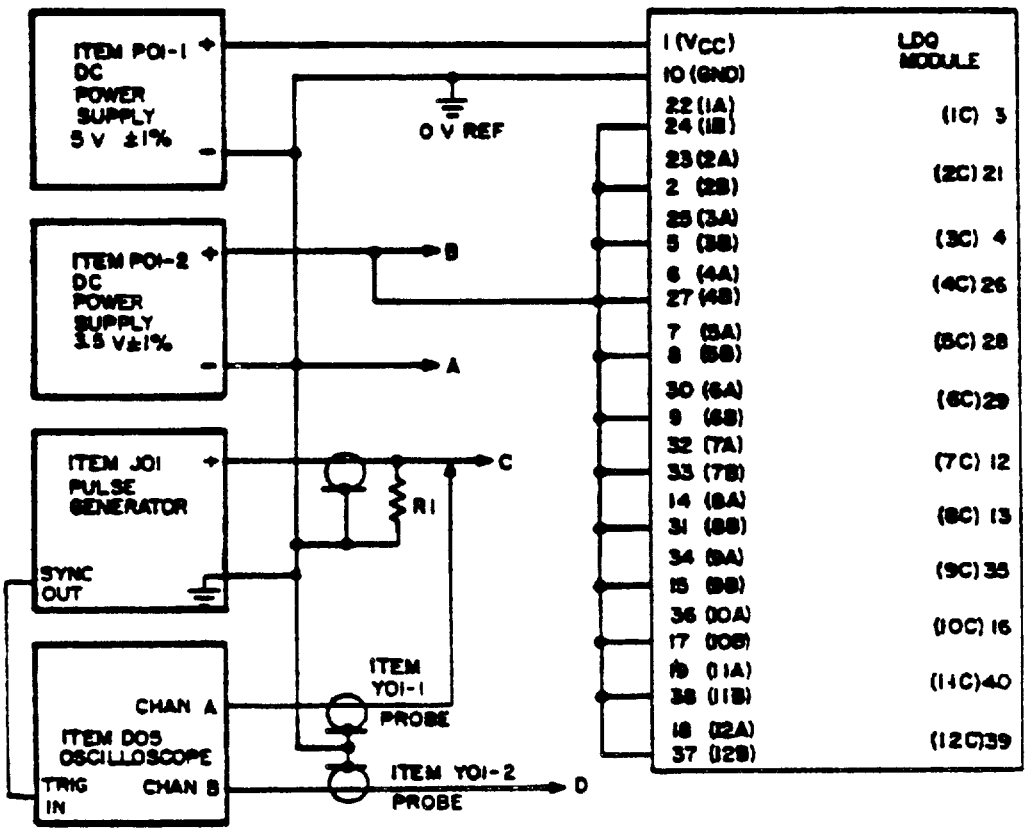
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P15	3C Output crosstalk, low level	Open	6,22,23,25	7,14,18,19,30, 32,34,36	4	0	280	mV	13
P16	4C "	"	6,22,23,25	7,14,18,19,30, 32,34,36	26	"	"	"	"
P17	5C "	"	7,14,30,32	6,18,19,22,23, 25,34,36	28	"	"	"	"
P18	6C "	"	"	"	29	"	"	"	"
P19	7C "	"	"	"	12	"	"	"	"
P20	8C "	"	"	"	13	"	"	"	"
P21	9C "	"	18,19,34,36	6,7,14,22,23, 25,30,32	35	"	"	"	"
P22	10C "	"	"	"	16	"	"	"	"
P23	11C "	"	"	"	40	"	"	"	"
P24	12C "	"	"	"	39	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 20260-20T15:28Z
Check the source to verify that this is the current version before use.

TABLE XV. Output crosstalk (high-level) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R1	1C Output crosstalk, high level	22	Open	6,7,14,18,19,23, 25,30,32,34,36	3	0	400	mV	13
R2	2C "	23	"	6,7,14,18,19,22, 25,30,32,34,36	21	"	"	"	"
R3	3C "	25	"	6,7,14,18,19,22, 23,30,32,34,36	4	"	"	"	"
R4	4C "	6	"	7,14,18,19,22,23, 25,30,32,34,36	26	"	"	"	"
R5	5C "	7	"	6,14,18,19,22,23, 25,30,32,34,36	28	"	"	"	"
R6	6C "	30	"	6,7,14,18,19,22, 23,25,32,34,36	29	"	"	"	"
R7	7C "	32	"	6,7,14,18,19,22, 23,25,30,34,36	12	"	"	"	"
R8	8C "	14	"	6,7,18,19,22,23, 25,30,32,34,36	13	"	"	"	"
R9	9C "	34	"	6,7,14,18,19,22, 23,25,30,32,36	35	"	"	"	"
R10	10C "	36	"	6,7,14,18,19,22, 23,25,30,32,34	16	"	"	"	"
R11	11C "	19	"	6,7,14,18,22,23, 25,30,32,34,36	40	"	"	"	"
R12	12C "	18	"	6,7,14,19,22,23, 25,30,32,34,36	39	"	"	"	"

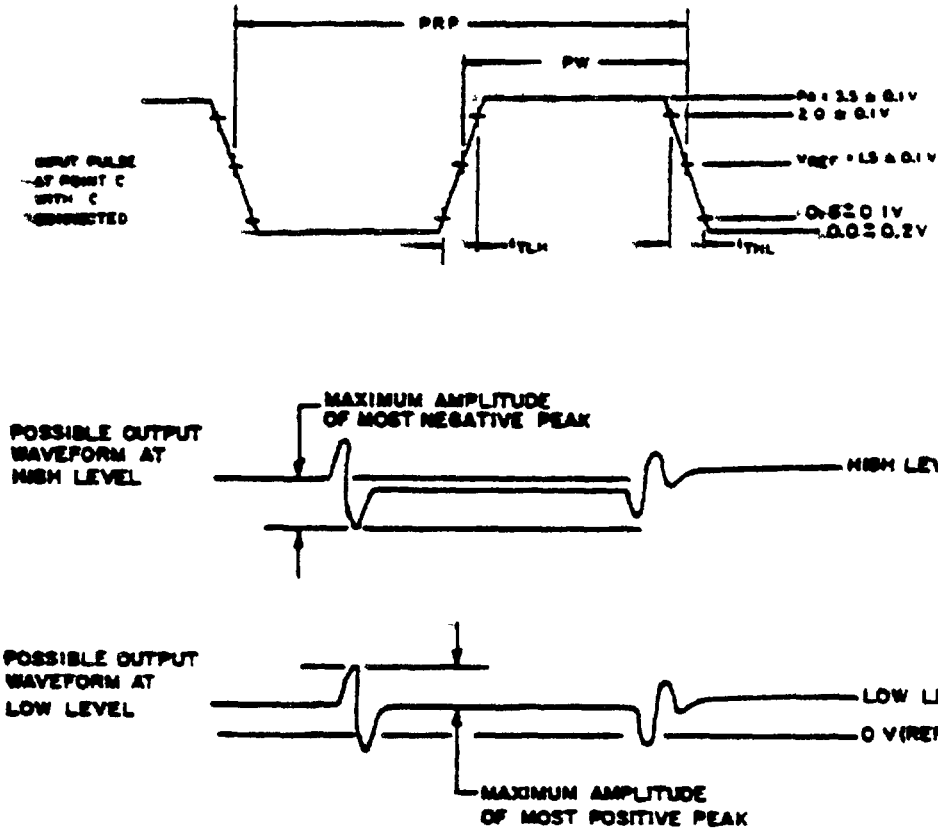
Source: https://assist.dla.mil -- Downloaded: 20260-20T15:28Z
Check the source to verify that this is the current version before use.



NOTES
1. R1 = 51Ω ±5%, 1 W.
2. Load each output with a load as specified on figure 9

FIGURE 13. Output crosstalk (low-level and high-level) test circuit and waveforms for tables XIV and XV

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:28Z
Check the source to verify that this is the current version before use.



NOTES.
1. Input pulse parameters:
   PWA = 200 ns ±10%.
   $t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input pin).
   PRP = 1 µs ±10%.
2. All voltage levels are referenced to ground.

FIGURE 13. Output crosstalk (low-level and high-level) test circuit and waveforms for tables XIV and XV - Continued.

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:28Z
Check the source to verify that this is the current version before use.

TABLE XVI. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
S	Power supply transient amplitude	0	400	mV	14

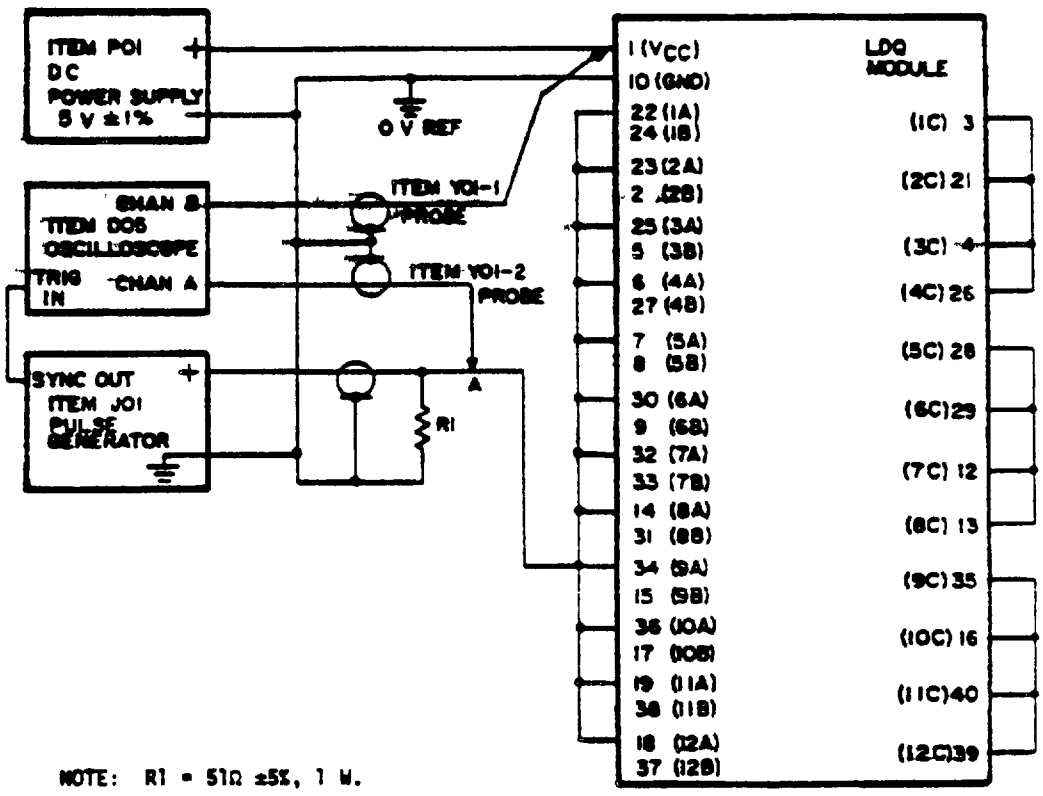
TABLE XVII. Power supply transient rise time test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
T	Power supply transient rise time	1	2	ms	15

TABLE XVIII. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
U1	Isolation	C,11,20	Frame	10	-	MΩ	16
U2	Isolation, unused pins	C	11,20	"	"	"	"
U3	Isolation, unused pins	11	20	"	"	"	"

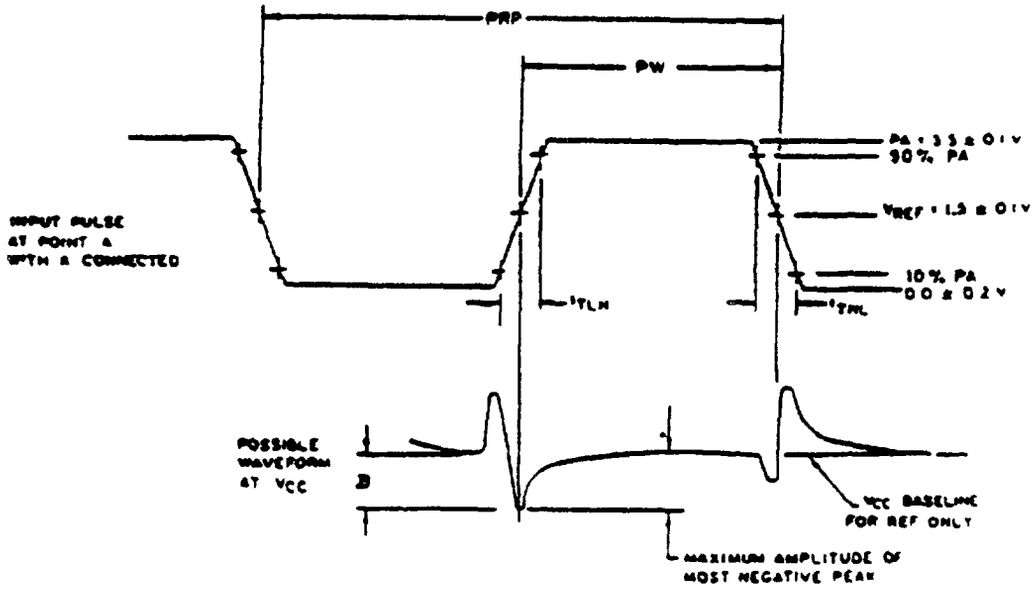
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:28Z
Check the source to verify that this is the current version before use.



NOTE: R1 = 51Ω ±5%, 1 W.

FIGURE 14. Power supply transient amplitude test circuit and waveforms for table XVI

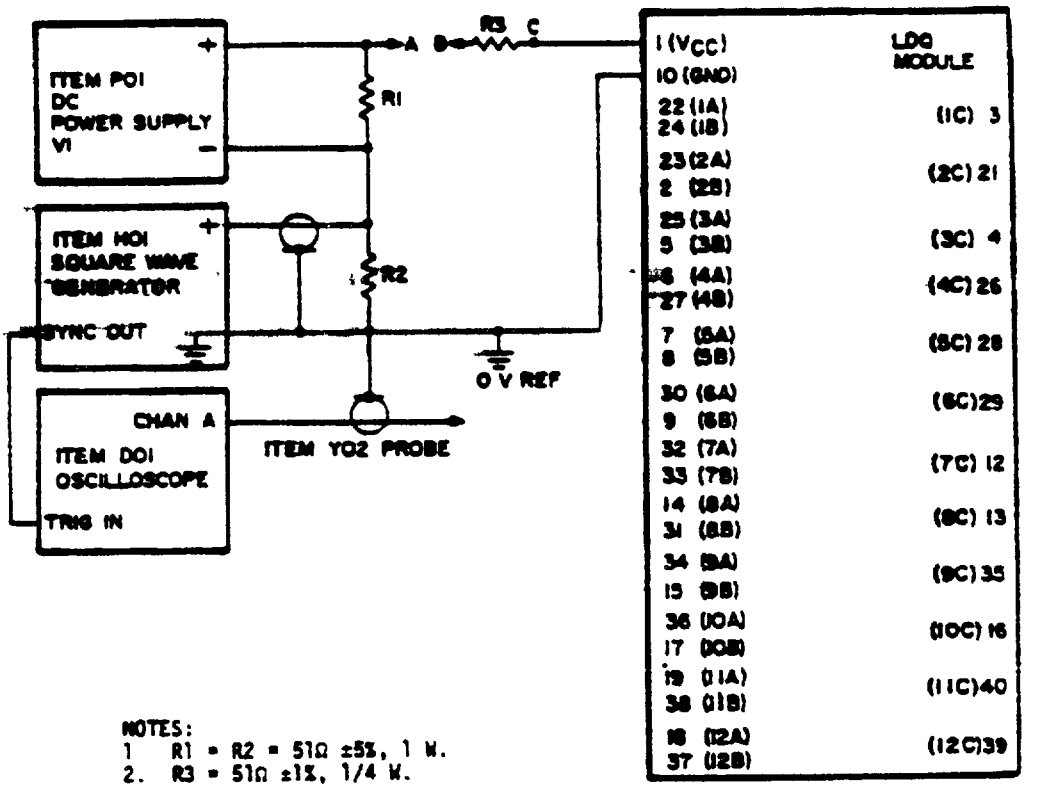
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:28Z
Check the source to verify that this is the current version before use.



NOTES:
1. Input pulse parameters:
   PWA = 3 µs ±10%.
   $t_{TLH} = t_{THL} = 10 \pm 1$ ns.
   PRP = 10 µs ±10%.
2. B = Measurement from the baseline to the most negative excursion of the waveform.
3. All voltage levels are referenced to ground.

FIGURE 14. Power supply transient amplitude test circuit and waveforms for table XVI - Continued.

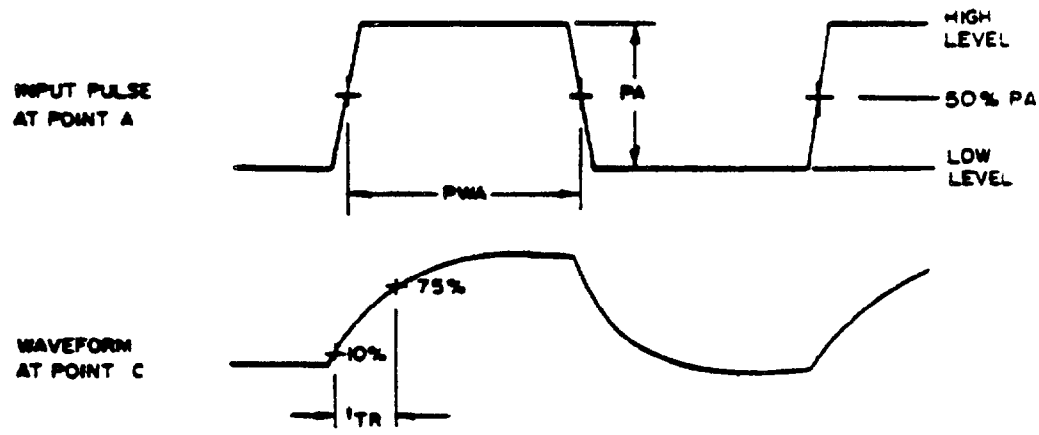
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:28Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = R2 = 51Ω ±5%, 1 W.
2. R3 = 51Ω ±1%, 1/4 W.

FIGURE 15 Power supply transient rise time test circuit and waveforms for table XVII

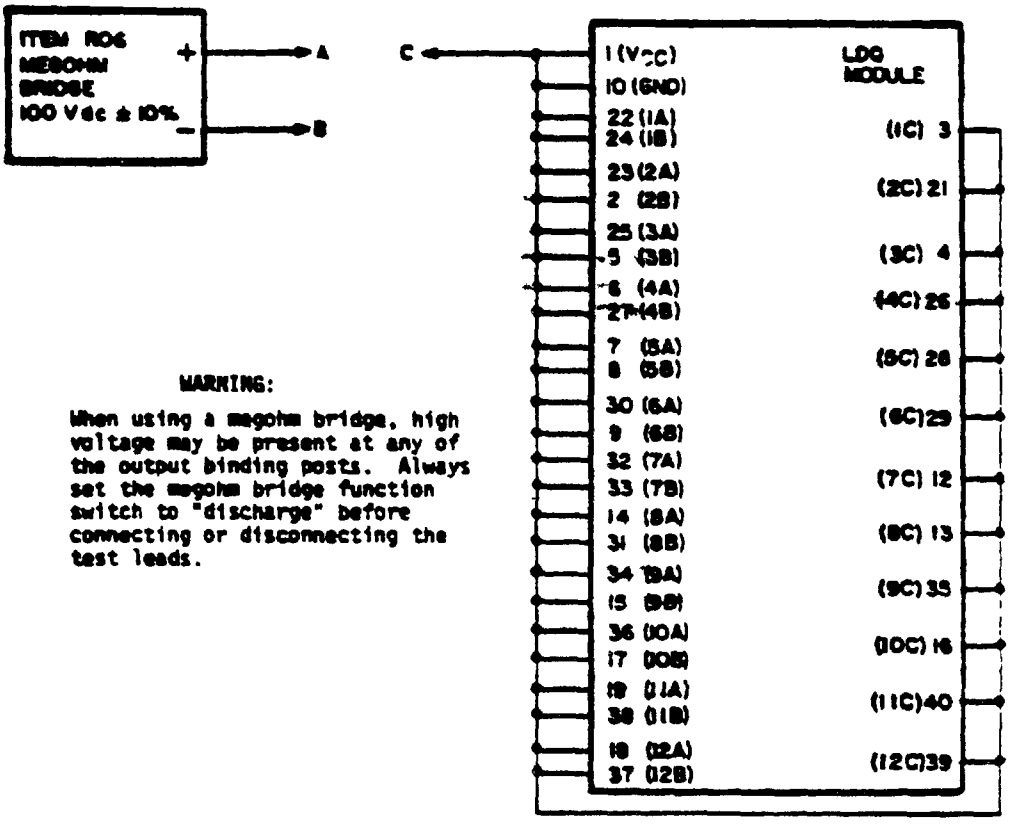
Source: https://assist.dla.mil -- Downloaded: 202610-20T15:28Z
Check the source to verify that this is the current version before use.



NOTES.
1. Input pulse parameters
   PA = 0.5 V_{p-p} ±1%.
   PWA = 10 ms ±10%.
   PRR = 50 Hz (reference).
   t_{TLH} and t_{THL} of input < 100 ns (reference).
2. With point "A" open, adjust input as shown.
3. With point "A" open, adjust V1 such that the low level of the input is at 0.00 ±0.1 V.
4. Connect "A" to "B", recheck the low level and pulse parameters at point "A"
5. Measure transient rise time (t_{TR}) at point "C"

FIGURE 15. Power supply transient rise time, test circuit and waveforms for table XVII - Continued

Source: https://assist.dla.mil -- Downloaded: 20260-20T15:28Z
Check the source to verify that this is the current version before use.



WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: For test condition U1 penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 16. Isolation test circuit for table XVIII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:28Z
Check the source to verify that this is the current version before use.

TABLE XIX. Input capacitance test.

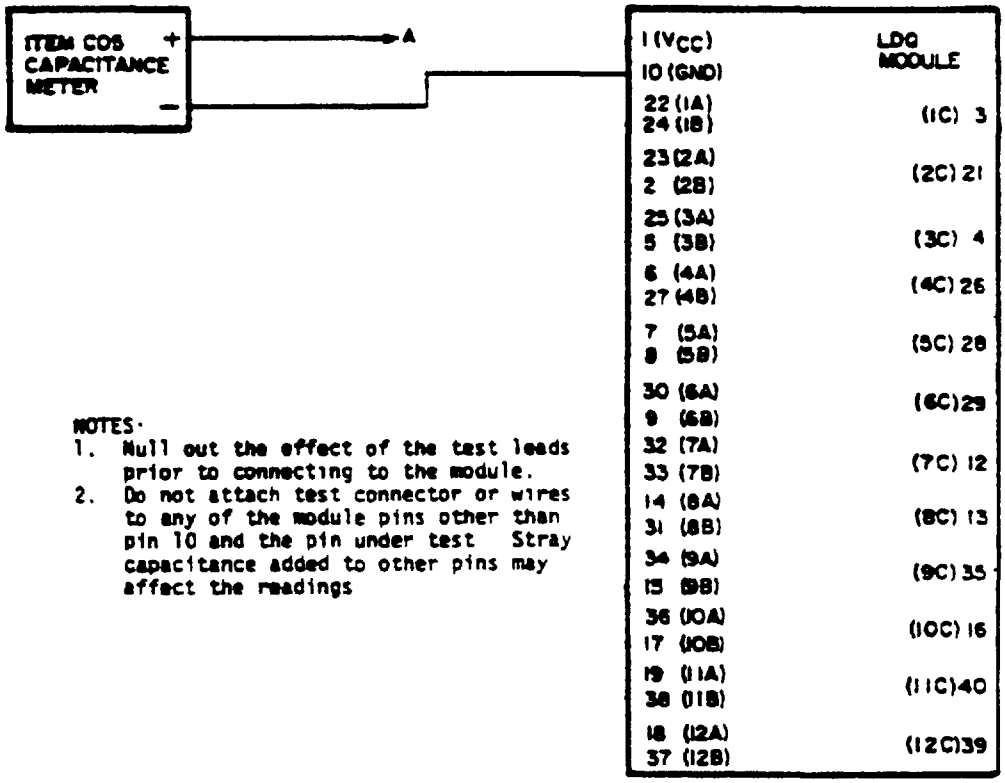
Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
V1	1A Input capacitance	22	2	15	pF	17
V2	1B "	24	"	"	"	"
V3	2B "	2	"	"	"	"
V4	2A "	23	"	"	"	"
V5	3B "	5	"	"	"	"
V6	3A "	25	"	"	"	"
V7	4A "	6	"	"	"	"
V8	4B "	27	"	"	"	"
V9	5A "	7	"	"	"	"
V10	5B "	8	"	"	"	"
V11	6B "	9	"	"	"	"
V12	6A "	30	"	"	"	"
V13	7A "	32	"	"	"	"
V14	7B "	33	"	"	"	"
V15	8A "	14	"	"	"	"
V16	8B "	31	"	"	"	"
V17	9B "	15	"	"	"	"
V18	9A "	34	"	"	"	"
V19	10B "	17	"	"	"	"
V20	10A "	36	"	"	"	"
V21	11A "	19	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:28Z
Check the source to verify that this is the current version before use.

TABLE XIX. Input capacitance test - Continued.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
Y22	11B Input capacitance	38	2	15	pF	17
Y23	12A "	18	"	"	"	"
Y24	12B "	37	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:28Z
Check the source to verify that this is the current version before use.



NOTES:
1. Null out the effect of the test leads prior to connecting to the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.

FIGURE 17 Input capacitance test circuit for table XIX

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:28Z
Check the source to verify that this is the current version before use.

TABLE XX. Output impedance (power off) check test.

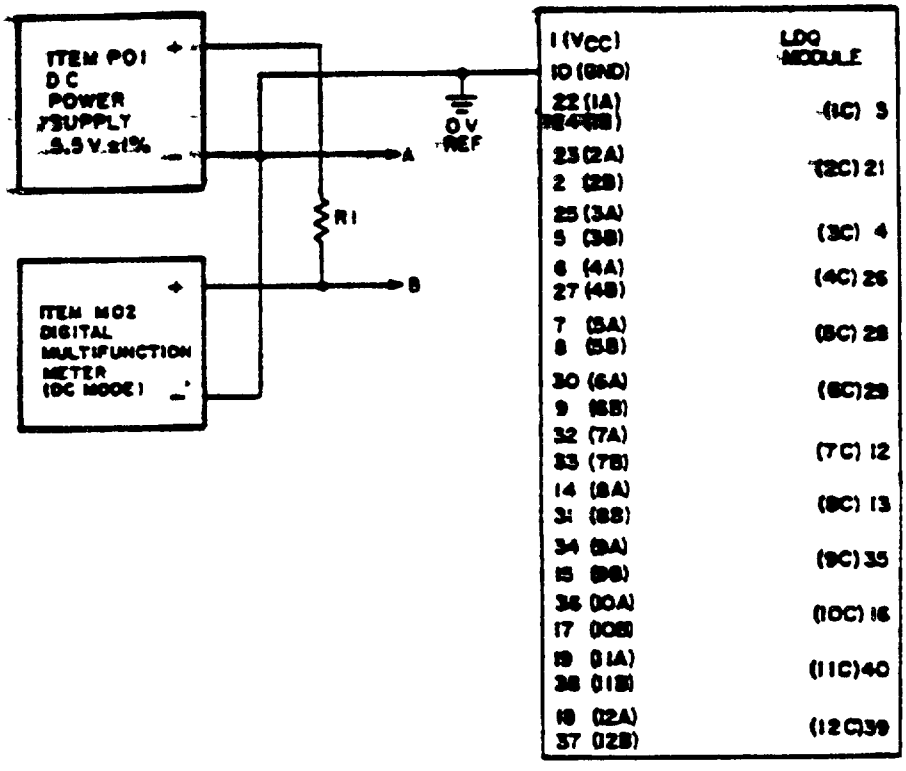
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
M1	1C Output impedance, power off, check	Open	3	3.50	5.55	V	18
M2	2C "	"	21	"	"	"	"
M3	3C "	"	4	"	"	"	"
M4	4C "	"	26	"	"	"	"
M5	5C "	"	28	"	"	"	"
M6	6C "	"	29	"	"	"	"
M7	7C "	"	12	"	"	"	"
M8	8C "	"	13	"	"	"	"
M9	9C "	"	35	"	"	"	"
M10	10C "	"	16	"	"	"	"
M11	11C "	"	40	"	"	"	"
M12	12C "	"	39	"	"	"	"
M13	1C "	1	3	"	"	"	"
M14	2C "	"	21	"	"	"	"
M15	3C "	"	4	"	"	"	"
M16	4C "	"	26	"	"	"	"
M17	5C "	"	28	"	"	"	"
M18	6C "	"	29	"	"	"	"
M19	7C "	"	12	"	"	"	"
M20	8C "	"	13	"	"	"	"
M21	9C "	"	35	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:28Z
Check the source to verify that this is the current version before use.

TABLE XX. Output impedance (power off) check test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
W22	10C Output impedance, power off, check	1	16	3.50	5.55	V	18
W23	11C "	"	40	"	"	"	"
W24	12C "	"	39	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:28Z
Check the source to verify that this is the current version before use.



NOTE: R1 = 10 kΩ ±1%, 1/8 W.

FIGURE 18. Output impedance (power off) check test circuit for table XX

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:28Z
Check the source to verify that this is the current version before use.

FOLD

Naval Electronic Systems Command
Washington, DC 20363

POSTAGE AND FEES PAID
DEFENSE LOGISTICS AGENCY
DoD-SCA



OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commander
Naval Electronic Systems Command
Defense Standardization Program Branch
Department of the Navy
Washington, DC 20363

ATTN: ELEX 8111

FOLD

Source: https://assist.dla.mil -- Downloaded: 202610-20T15:28Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS. This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER (Number) AND TITLE

MIL-M-28787/18B(EC) MODULES, STANDARD ELECTRONIC, GATE, NAND, DIGITAL, KEY CODE CDQ

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1 ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C REASON FOR RECOMMENDED CHANGE(S)

2 REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO

DATE

DD FORM 1 OCT 76 1426

EDITION OF 1 JAN 72 MAY BE USED UNTIL EXHAUSTED

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:28Z
Check the source to verify that this is the current version before use.

INCH-POUND

MIL-M-28787/18B(NW)
AMENDMENT 2
30 July 1993
SUPERSEDING
AMENDMENT 1
29 June 1990

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

GATE, NAND, DIGITAL, KEY CODE LDQ

This amendment forms a part of Military Specification MIL-M-28787/18B(EC), dated 27 April 1983, and is approved for use by the Crane Division, Naval Surface Warfare Center, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 4

TABLE I, Test conditions H1-H12, Test type column: Delete "X" and substitute "S".

PAGE 5

*TABLE I, Test conditions U2,U3, Test type column: Delete "Q" and substitute "S".

TABLE I, Test conditions V1-V24, Max columns: Delete "15" (five places) and substitute "19" (five places).

PAGE 6

4.4.2: Delete and substitute the following:

"4.4.2 Order of testing. Test conditions A through W shall be performed in any order with the following exceptions. Test conditions A, B, C, H, and U are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, B, C, H, or U are part of a sequence of tests, the order of testing shall be as follows: U, A, B, C, and H where applicable, followed by the remaining tests in the sequence."

4.5.1, fifth line: Delete "C through N and T" and substitute "D through G, J through N and T".

4.5.3, third line: Delete "C" and substitute "D through G, J".

4.5.3, fifth and sixth lines: Delete "C through W" and substitute "D through G, J through N, and T".

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:28Z
Check the source to verify that this is the current version before use.